

1 Agroforestry SmartAgroforst: Poplar biomass modelling

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3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

6 1. Introduction

7 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
8 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
9 biodiversity conservation, and rural income generation on the same hectare of land [14].

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10 Temperate agroforestry systems (AFS) based on fast-growing species such as poplar, wil-
11 low, alder, and black locust can deliver substantial biomass yields while improving soil
12 quality and providing multiple ecosystem services compared to conventional monocul-
13 tures [7]. Biomasses produced in AFS is considered as a sustainable feedstock and can
14 be used either energetically, chemically or materially (e.g. in the building sector). The
15 latter two options are generally preferred over the energetic usage from a cleaner pro-
16 duction perspective as biogenic carbon is stored in products during their life span which
17 contributes to climate change mitigation.

18 Despite substantial research on ecological advantages and pioneering initiatives to
19 establish AFS in practice, the share of AFS in agricultural production systems is still
20 small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025,
21 which accounts for about 0.02% of total agricultural land use [DEFAP]. One of the core
22 challenges in establishing AFS systems in large scale in practice, is a lack of opportunities
23 to generate substantial revenues from woody biomasses of AFS. On the other side, the
24 rise of bioeconomy creating substantial demands of woody biomass requires a reliable
25 and cost-efficient feedstock supply potentials. As woody biomasses are bulky feedstocks
26 with low energy density, the design of cost-efficient supply chains is crucial to ensure
27 the economic viability of AFS-based value chains. This requires coordinated decisions
28 on stand establishment, multi-cycle harvesting, and product cascading in spatially dis-
29 tributed supply chains, which can be supported by integrated optimisation models that
30 link stand-level growth dynamics to regional supply chain design.

31 In the last two decades, a large body of research has analysed biomass supply chains
32 for bioenergy and biorefineries using mathematical programming models, often in the
33 form of mixed-integer linear programming (MILP) [16, 12]. These models typically opti-
34 mise network configuration, feedstock sourcing, and logistics under cost or profit objec-
35 tives. They often treat biomass availability as an exogenous input or rely on simplified
36 agricultural production decisions. Moreover, downstream processing of biomasses is con-
37 centrated on one particular value chain e.g. bioenergy plants consuming all produced
38 biomasses. Recent work has started to integrate operational details such as various
39 biomass types, biomass growth models and regeneration processes in multi-period sup-
40 ply chain models.

41 The present paper contributes to this literature in four ways. First, we propose
 42 a novel MILP formulation for AFS-based biomass supply chain design that explicitly
 43 links establishment decisions and consecutive harvest decisions, ensuring that harvesting
 44 decisions at each site are consistent with minimum and maximum rotation ages. Second,
 45 we embed allometric yield functions for different biomass types generating age-specific
 46 yields for different woody biomass types (like stems, branches, or leaves and twigs). Third,
 47 these different woody biomass types can be processed in specific downstream industries,
 48 like chemical or paper industry as well as energy sector. The model aims to provide
 49 strategic decision support for establishing regional AFS-based bioeconomy ecosystems
 50 connecting multiple industries and product markets.

51 The remainder of the paper is structured as follows. Section @ref(sec:litreview) re-
 52 views the relevant literature on agroforestry biomass systems, allometric biomass mod-
 53 elling, and biomass supply chain optimisation. Section @ref(sec:allometrics) introduces
 54 the allometric models adopted to represent biomass growth of different tree species and
 55 biomass types. Section @ref(sec:milp) summarises the MILP formulation for the agro-
 56 forestry supply chain with age-lag constraints. Section @ref(sec:experiments) outlines the
 57 planned computational experiments for the case study and discusses expected insights
 58 for cleaner production strategies.

59 **2. Literature review**

60 Agroforestry biomass systems, tree and stand biomass models, and biomass sup-
 61 ply chain optimisation have all been studied extensively, but they are rarely combined
 62 in an integrated modelling framework that links age-dependent stand dynamics to re-
 63 gional product-cascading value chains [16, 12, 5]. This section reviews (i) agroforestry
 64 and short-rotation systems with emphasis on yields and economics, (ii) allometric and
 65 age-dependent biomass yield models, and (iii) biomass supply chain optimisation and
 66 cascading use of biomass, and concludes by identifying the modelling gap addressed in
 67 this paper.

68 *2.1. Biomass supply chain optimisation*

69 Biomass supply chain design and planning have been widely studied using MILP
 70 models at strategic, tactical, and operational levels Zahraee et al. [16]; Sharma et al.

[12]; Espinoza et al. [5]. Comprehensive reviews classify models according to decision scope (location, capacity, routing, inventory), time horizon, uncertainty treatment, and sustainability metrics, and show a strong emphasis on bioenergy and biorefinery systems Lamers et al. [8]; Espinoza et al. [5]. Typical models optimise the number, size, and locations of processing facilities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost- or profit-oriented objectives, sometimes complemented by greenhouse gas emissions or other environmental indicators Eksioglu et al. [4]; Arabi et al. [1].

A key methodological feature of many biomass supply chain models is the integration of long-term location and capacity decisions with multi-period logistics decisions such as harvesting, transport, storage, and processing Eksioglu et al. [4]; Zahraee et al. [16]. For example, mixed-integer formulations have been used to simultaneously determine biorefinery locations, their capacities, biomass collection and storage sites, and time-dependent flows of biomass and biofuel, explicitly accounting for seasonality, deterioration, and inventory trade-offs Eksioglu et al. [4]. These models demonstrate that a significant share of biofuel production costs is attributable to logistics, and that integrated optimisation of network design and operations can substantially reduce costs compared to decoupled decision making Eksioglu et al. [4]; Arabi et al. [1].

However, in virtually all of these models, biomass availability over time is treated as an exogenous input, either as annual supply potentials per region or as simple time profiles based on harvest windows and assumed yields Zahraee et al. [16]; De Meyer et al. [3]. Stand-level growth dynamics, age-class structure, and rotation decisions are not represented explicitly; instead, land-use or yield scenarios enter as fixed parameters De Meyer et al. [3]; Lamers et al. [8]. As a result, these models cannot explore trade-offs between rotation length, stand management, and supply chain configuration in a unified optimisation framework, especially not for agroforestry systems with repeated coppice cycles and strong interactions between tree growth and harvesting decisions.

2.2. *Agroforestry and short-rotation systems*

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance Nair [10]; Toensmeier [14]. In temperate regions, short-rotation

102 agroforestry systems (SRAFS) often take the form of alley-cropping, where double or
 103 multiple tree rows alternate with crop strips, enabling simultaneous production of agri-
 104 cultural and woody biomass Huber et al. [7]. Empirical trials in Southern Germany and
 105 Central Europe show that SRAFS with poplar, willow, black alder, and black locust can
 106 achieve mean annual increments (MAI) of roughly 7–10 t per ha and year in aboveground
 107 woody biomass during the first rotation, with pronounced species-specific differences in
 108 growth trajectories, stand structure, and mortality (see Huber et al. [7]; Trnka et al.
 109 [15]).

110 For short-rotation coppice (SRC) and very short-rotation coppice (vSRC) poplar plan-
 111 tations in Italy and Central Europe, economic analyses indicate that rotations of about
 112 5–7 years at moderate planting densities can be competitive with arable crops, but prof-
 113 itability is highly sensitive to yield assumptions, establishment costs, and logistics [13].
 114 This sensitivity implies to couple stand-level growth and models with supply chain optimi-
 115 sation models, rather than treating biomass availability as an exogenous, time-invariant
 116 parameter [16]. Beyond yields and economics, agroforestry and SRC systems have been
 117 widely analysed for carbon sequestration, nutrient cycling, and biodiversity, reinforcing
 118 their role as candidate feedstock systems for regional bioeconomy strategies that empha-
 119 sise material-before-energy use of biomass (see e.g. Toensmeier [14]).

120 Thus, a fundamental cornerstone of AFS modeling is the representation of biomass
 121 growth dynamics, which can be achieved through allometric models that link tree bio-
 122 metric variables to biomass, and through age-dependent stand biomass functions that
 123 capture growth trajectories over rotation cycles. These models provide the necessary
 124 link between stand structure and biomass availability, which can then be integrated into
 125 supply chain optimisation models to explore trade-offs between rotation length, product
 126 portfolios, and infrastructure configuration in AFS supply chains.

127 *2.3. Tree-level allometric models*

128 Allometric tree biomass models relate easily measured variables, such as stem base di-
 129 ameter (SBD) or diameter at breast height, to aboveground dry biomass using power-law
 130 functions of the form $M = b_0 \text{SBD}^{b_1}$, with species-specific coefficients b_0 and exponents b_1
 131 [6]. These models allow non-destructive estimation of individual-tree and stand biomass

132 from diameter measurements and have been widely used in forestry and agroforestry for
133 both carbon accounting and management planning [6, 14].

134 For SRAFS in Southern Germany, species-specific allometric functions based on SBD
135 measured 10 cm above ground were developed for black alder, black locust, poplar (Max
136 3), and willow (Inger) Huber et al. [6]. The reported models share a common exponent
137 $b_1 = 2.603$ across species, while the scale parameter b_0 ranges from about 0.025 to 0.041,
138 reflecting differences in wood density and branching pattern Huber et al. [6]. Applying
139 these functions to all shoots per tree and aggregating over observed tree densities yields
140 annual aboveground leafless dry biomass at plot and stand level, from which current
141 annual increment (CAI) and MAI curves can be derived for each species and management
142 system Huber et al. [7].

143 These SRAFS studies demonstrate that stand structure, diameter distributions, and
144 mortality strongly affect aggregate yields, with black locust showing high wood density
145 and strong size inequality and willow showing high shoot densities but low wood density
146 and diameter growth Huber et al. [7]. From a modelling perspective, the key point is
147 that tree-level allometrics provide a robust mapping from biometric state variables to
148 biomass, which can be exploited either to build detailed stand simulators or to construct
149 age-class yield tables for optimisation models Huber et al. [6].

150 *2.4. Age-dependent biomass yield functions*

151 Existing allometric models link tree biomass to diameter or height but do not yet
152 provide **stand-level** biomass as an explicit function of stand age. Age-dependent biomass
153 functions are useful for yield projections in short-rotation forestry and agroforestry, where
154 management is organized in discrete rotations (e.g. 5–15 years). [11, 13]

155 A common approach is to derive age functions from rotation trials that report mean
156 annual increment (MAI) over time and identify the age at which MAI plateaus. For
157 poplar SRWC in northern Poland, Niemczyk et al. (2021) showed that MAI in above-
158 ground dry biomass increases up to about age 10; beyond this, the MAI curve flattens,
159 suggesting a biologically optimal rotation age around 10 years because the slope of MAI
160 is near zero. [11] Reported MAI values for single-stem poplar plantations in a 10-year
161 rotation ranged approximately from 1–15 Mg DM ha⁻¹ yr⁻¹ depending on cultivar. [11]
162 Similar magnitudes and rotation lengths (about 5–7 years for SRC, up to 10–15 years for

163 SRC/SRF) are reported for poplar SRC systems in Italy, where whole-rotation averages
 164 of roughly 15 Mg DM ha⁻¹ yr⁻¹ are considered economically viable. [13]

165 A simple and widely used parametric form for age-dependent stand biomass $B(t)$ is
 166 the Chapman–Richards (or Mitscherlich/monomolecular) function:

$$B(t) = B_{\max} (1 - e^{-k \cdot t})^m$$

167 with stand age t in years, asymptotic biomass $B_{\max}$ (Mg DM ha⁻¹), rate parameter k
 168 (year⁻¹), and shape parameter m . This formulation captures the rapid juvenile growth,
 169 a phase of approximately linear biomass accumulation, and a saturating phase where
 170 growth slows. [11] The corresponding current annual increment (CAI) and mean annual
 171 increment (MAI) are

$$\text{CAI}(t) = \frac{dB}{dt}, \quad \text{MAI}(t) = \frac{B(t)}{t}.$$

172 The age maximizing MAI (optimal rotation under a volume/biomass criterion) can be
 173 derived analytically or determined numerically. Fitting such functions to observed age–
 174 biomass or age–volume series from rotation trials allows cultivar-specific yield curves that
 175 are directly usable in agroforestry simulation models. [11, 13]

176 For illustration, we parameterize three stylized poplar curves representing low, medium,
 177 and high productivity on temperate mineral soils. Parameter values are loosely informed
 178 by the ranges reported by Niemczyk et al. (2021) and Testa et al. (2014) and should be
 179 recalibrated when site-specific data are available. [11, 13]

- 180 • Low productivity: $B_{\max} = 80$ Mg DM ha⁻¹, $k = 0.18$, $m = 1.1$
- 181 • Medium productivity: $B_{\max} = 120$ Mg DM ha⁻¹, $k = 0.20$, $m = 1.2$
- 182 • High productivity: $B_{\max} = 160$ Mg DM ha⁻¹, $k = 0.22$, $m = 1.3$

183 These values yield MAI maxima between about 8 and 12 years with peak MAI values
 184 of roughly 8–16 Mg DM ha⁻¹ yr⁻¹, consistent with the empirical range where 10-year
 185 rotations produced MAI of about 10–15 Mg DM ha⁻¹ yr⁻¹ for the best poplar cultivars
 186 in northern Poland and Italy. [11, 13]

187 In the agroforestry context, such age functions can be combined with tree density and
 188 allometric partitioning to above- and below-ground biomass to obtain tree-row biomass

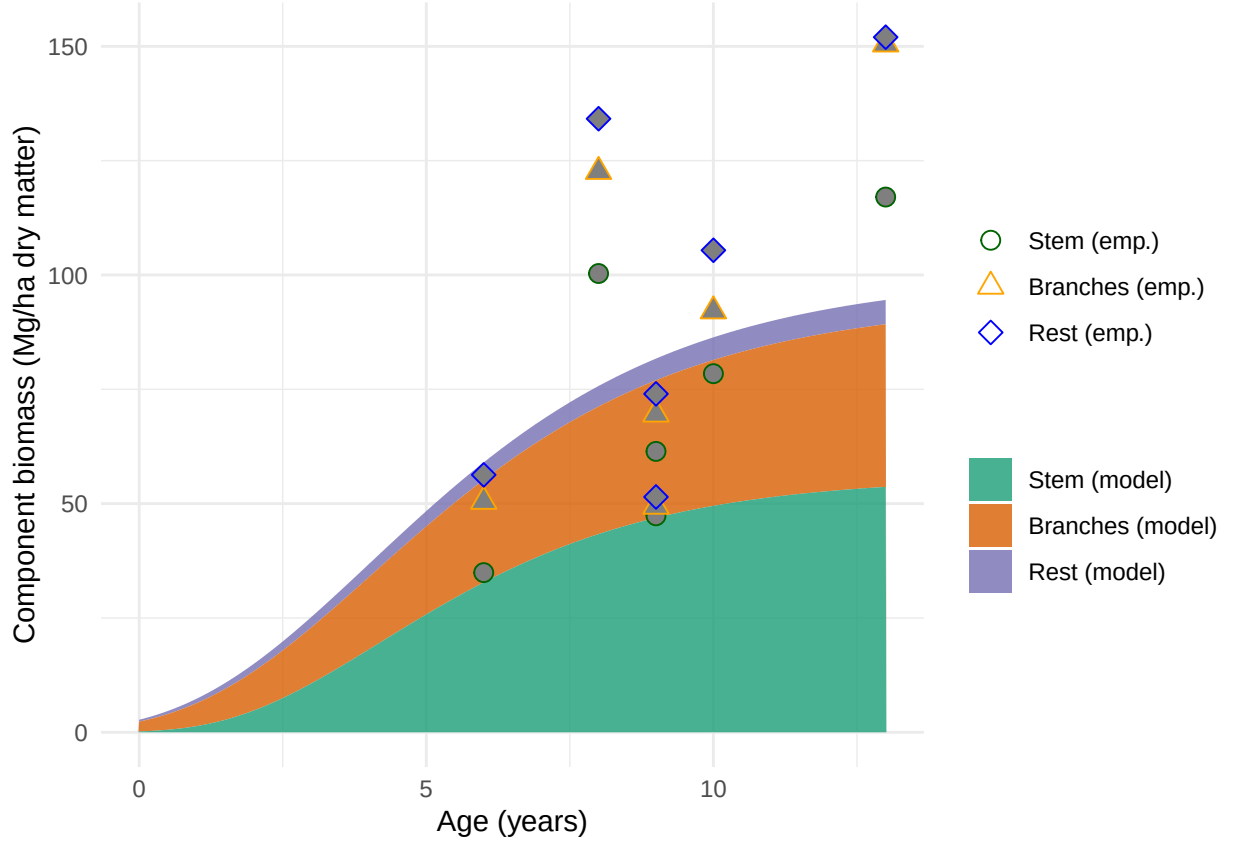


Figure 1: Modeled stand-level biomass components for poplar clone Max 3 over age, represented as stacked Gompertz curves and compared with empirical reference values from agroforestry trials.

over time for different designs. Calibration against local inventory data or growth measurements (e.g. basal area increment and height growth) is recommended before using the curves for scenario analysis.

2.5. Cascading use of biomass and product quality

From a cleaner production perspective, cascading use of biomass—prioritising high-value material uses before energy recovery—is widely advocated to maximise value creation and reduce environmental burdens Lamers et al. [8]; Espinoza et al. [5]. In wood-based value chains, this typically implies allocating higher-quality stem fractions to products such as sawn timber, veneer, panels, or pulp, while directing lower-quality fractions and residues to energy applications De Meyer et al. [2]. Multi-product MILP models such as

199 OPTIMASS implement this concept by defining a hierarchy of product quality classes
200 and allowing higher-quality products to satisfy lower-quality demands via substitution
201 constraints along the cascade De Meyer et al. [2]; De Meyer et al. [3].

202 For agroforestry biomass such as poplar and willow, this logic translates into a set of
203 product streams (e.g. larger-diameter logs for material uses, medium-diameter fractions
204 for pulp and panels, small-diameter wood and residues for energy), whose relative shares
205 depend on stand age and growth conditions Huber et al. [7]; Trnka et al. [15]. In the
206 present work, the age-class yield construction described above is extended by associating
207 each age class with product-specific yield fractions, thereby capturing age-dependent
208 quality in a simplified but optimisation-compatible form. The MILP formulation then
209 enforces a product hierarchy in which higher-quality products can satisfy lower-quality
210 demands, subject to the available yields and infrastructure capacities, thus embedding a
211 simplified cascading logic into the regional supply chain design De Meyer et al. [2].

212 *2.6. Synthesis and modelling gap*

213 The literature summarised above provides three critical ingredients for modelling agro-
214 forestry biomass supply chains. First, empirical SRAFS and SRC studies offer species-
215 and site-specific information on growth, MAI, stand structure, and economic performance,
216 including allometric models that link diameter distributions to biomass Huber et al. [6];
217 Huber et al. [7]; Trnka et al. [15]; Testa et al. [13]. Second, age-dependent stand biomass
218 functions such as Chapman–Richards provide a convenient parametric representation of
219 growth over rotation, from which discrete age-class yields can be derived Niemczyk et al.
220 [11]. Third, biomass supply chain MILP models and multi-product cascading formula-
221 tions offer a rich toolkit for optimising infrastructure, logistics, and product allocation
222 in regional bioeconomy systems Zahraee et al. [16]; De Meyer et al. [2]; De Meyer et al.
223 [3]; Espinoza et al. [5].

224 At the same time, existing models fall short of explicitly coupling agroforestry stand
225 dynamics with supply chain design in several respects. Most biomass supply chain for-
226 mulations assume exogenous biomass availability profiles and do not represent stand age
227 or rotation decisions, while forestry-oriented growth and yield models are rarely embed-
228 ded in regional optimisation models that include multiple products and cascading uses
229 De Meyer et al. [3]; Lamers et al. [8]. In particular, there is a lack of models that (i)

link age-dependent stand biomass functions to age-class yield tables for different product types, (ii) enforce multi-cycle age-lag constraints that ensure consistency of harvesting decisions at each agroforestry site across successive rotations, and (iii) allocate biomass along a product quality cascade within a spatially explicit, multi-period supply chain optimisation framework De Meyer et al. [2]; De Meyer et al. [3]; Eksioglu et al. [4].

The MILP formulation developed in this paper addresses this gap by integrating age-dependent biomass yield modelling, stand-level age-lag constraints, and a product-cascading hierarchy into a single optimisation model for the design and operation of agroforestry biomass supply chains.

3. Biomass supply chain modelling

Biomass supply chain design and planning has been widely studied using MILP models at strategic, tactical, and operational levels [16, 12]. Comprehensive reviews classify models according to decision scope (location, capacity, routing, inventory), time horizon, uncertainty treatment, and sustainability metrics [16, 5]. Many models focus on bioenergy supply chains for electricity, heat, or biofuels, considering feedstocks such as agricultural residues, forestry residues, dedicated energy crops, and municipal wastes [12]. Environmental aspects, including greenhouse gas emissions and land use, are increasingly integrated, often using life-cycle assessment (LCA) data or emission factors [5].

Among generic models, OPTIMASS is a notable example of a multi-product MILP that optimises biomass supply chains with changing biomass characteristics and by-product re-injection [2]. It models the flow of multiple biomass types and products through supply chain stages, capturing conversion yields, storage, and cascading uses. The t-OPTIMASS extension further incorporates biomass growth and regeneration over multiple periods, enabling the representation of time-dependent availability for biomass supply chains [3]. However, these formulations generally assume exogenous biomass availability profiles per period, and do not explicitly model stand-level age dynamics or rotation constraints.

Several scenario-based approaches have been developed for forest biorefinery supply chains, linking product portfolios, process choices, and supply chain design under market volatility [9]. Lamers et al. [8] discuss strategic feedstock supply system design for

biorefineries, emphasising the importance of economies of scale, conversion yields, and logistics in achieving cost-efficient and robust systems. Reviews highlight that only a limited subset of models explicitly consider biomass growth, spatially explicit land-use change, or agroforestry systems [16, 12]. There is therefore a clear opportunity to develop models that couple agroforestry stand dynamics with network design and cleaner production objectives.

3.1. Cascading use of biomass

From a cleaner production perspective, the cascading use of biomass—prioritising high-value material uses before energy recovery—is widely advocated to maximise resource efficiency and mitigate environmental impacts [8]. In wood-based value chains, this implies that high-quality timber or chemical feedstock fractions should first be allocated to material products such as engineered wood products, pulp and paper, or biochemicals, while lower-quality fractions and residues are used for energy [2, 8]. MILP models can represent such cascades by defining product quality classes and allowing substitution of higher-quality products for lower-quality demands, subject to hierarchy constraints.

The OPTIMASS framework explicitly models such product cascades by defining a product quality order and allowing flows from higher-quality products to lower-quality demand nodes [2]. Similar ideas have been applied in forest-based biorefinery supply chains, where product portfolios include platform chemicals, fuels, and electricity [9]. For agroforestry biomass, poplar and willow can supply different product streams, including sawlogs, peeled veneers, pulpwood, and energy chips, with quality primarily defined by diameter, straightness, and defect rates [13?]. Economic analyses in Italy and Central Europe suggest that the economic viability of SRC systems depends on access to higher-value markets such as panel products and pulp, in addition to energy markets [13].

Cleaner production strategies that integrate agroforestry biomass into regional bioeconomies therefore require models that can allocate biomass flows along such cascades while respecting stand growth dynamics, rotation constraints, and logistical capacities. Explicit representation of age-dependent yield quality (e.g., diameter distributions, wood density) remains challenging in large-scale MILP models, but can be approximated through age- and species-specific product yield fractions.

291 3.2. Stand-level yield functions and age classes

292 To integrate allometric growth into a MILP formulation, we abstract from individual
 293 trees and construct age-class yield functions for each biomass type p and age a . Following
 294 common practice in forest and SRC modelling, we define discrete annual age classes
 295 and compute the stand biomass yield Y_{pa} (t ha^{-1}) associated with harvesting at age a
 296 by combining allometric functions with empirically observed diameter distributions and
 297 stand densities [7, 15]. Where detailed stand structure data are available, this can be done
 298 by simulating SBD distributions over time (e.g., via mixed-effects models) and applying
 299 tree-level allometrics to each diameter class [7]. In the absence of such detailed data, age-
 300 dependent MAI and CAI curves reported in the literature can be used to approximate
 301 Y_{pa} [15, 7].

302 In the context of the agroforestry supply chain design (SCD) model, we assume
 303 that each agroforestry site i is characterised by a species or species mix, a planting
 304 density, and management regime, which jointly determine a site-specific yield profile η_{pa}
 305 (t ha^{-1}) for each product type p at age a . These yield profiles can be interpreted as the
 306 maximum harvestable biomass per hectare for product p if the stand is harvested at age
 307 a , conditional on the stand having been established at an earlier time. In the current
 308 MILP formulation, these profiles are captured through the parameter η_{pa} , termed base
 309 yield, and are linked to decision variables via the constraint

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i, p, t \in \mathcal{T}^{\text{harv}}.$$

310 Here, Y_{ipt} is the maximum harvest quantity of product p at site i in period t , AREA_i is
 311 the site area (ha), and z_{ist} is a binary variable indicating that harvests at site i in periods
 312 s and t form consecutive harvests with an age lag $t - s$ within allowed bounds. The term
 313 $\eta_{p(t-s)}$ thus plays the role of an allometrically derived yield coefficient that depends only
 314 on stand age $a = t - s$, not on calendar time.

315 3.3. Biomass types and product quality classes

316 In the model, three product types are distinguished: $p = 1$ (chemical-grade biomass),
 317 $p = 2$ (pulp-grade biomass), and $p = 3$ (energy-grade biomass). This tripartite classi-
 318 fication reflects a product quality hierarchy, with product 1 having the highest quality

and price, and product 3 being the lowest-quality fraction typically used for energy [2, 8]. From an allometric perspective, the differentiation among these products can be linked to diameter classes, stem straightness, and defect rates, as these determine the suitability of biomass fractions for different processing routes [13?].

In practice, allometric models provide estimates of total aboveground woody biomass, which can be split into product classes using empirical conversion factors or allocation rules derived from product recovery studies (e.g., share of biomass suitable for veneer vs. pulp vs. chips as a function of diameter and log length) [?]. For the purposes of the SCD model, we can represent this by specifying product-specific yield coefficients η_{pa} that sum to the total biomass yield at age a and reflect the expected proportion of each product class under a given management regime.

The cascading hierarchy in the MILP formulation is implemented through demand satisfaction constraints that allow higher-quality products to satisfy lower-quality demands. Specifically, demand $D_{kpt}^{\max}$ at consumer k for product class p in period t can be met by flows of product p and all higher-quality classes $p' \leq p$ from storage hubs, as captured by

$$\sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k, p, t.$$

This structure supports cleaner production by favouring material uses when possible and relegating residual fractions to energy uses, depending on revenue parameters $R_{k,p}$ and system constraints [2, 8].

4. MILP model for agroforestry supply chain design

4.1. Sets, indices, and time structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/processing facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and product types $\mathcal{P} = \{1, 2, 3\}$. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To represent stand establishment and harvest timing, extended time sets $\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$ and harvest periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$ are defined, where $A^{\min}$ and $A^{\max}$ are minimum and maximum stand ages (years) at harvest. The set $\mathcal{S}$ comprises all ordered pairs (s, t) of consecutive harvest periods (including

establishment at $s = 0$) satisfying the age-lag bounds:

$$\mathcal{S} = \{(s, t) \mid s, t \in \{0, \dots, T_{\max} + 1\}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}.$$

4.2. Decision variables

Binary variables z_{ist} indicate whether, at site i , periods s and t form consecutive harvests (or establishment and first harvest if $s = 0$), with $(s, t) \in \mathcal{S}$. These variables define a path of consecutive harvest cycles at each site over the planning horizon. Continuous variables $Y_{ipt} \geq 0$ represent the maximum harvest quantity (t) of product p at site i in harvest period t . Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from sites to storage and from storage to consumers, respectively, and S_{jpt} represent inventories at storage locations.

4.3. Objective function

The model maximises net present value over the planning horizon, expressed as total revenues minus establishment, opportunity, harvest, transport, and storage costs:

$$\begin{aligned} \max Z = & \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} \\ & - \sum_i C_i^{\text{est}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} z_{i0t} \\ & - \sum_i C_i^{\text{opp}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} (T_{\max} - t) \cdot z_{i0t} \\ & - \sum_{i, (s,t) \in \mathcal{S}: s > 0} C_i^{\text{harv}} \cdot \text{AREA}_i \cdot z_{ist} \\ & - c^{\text{tr-raw}} \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} - c^{\text{tr-pre}} \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} - \sum_{j,p,t} c_j^{\text{stor}} \cdot S_{jpt}. \end{aligned}$$

Here, $R_{k,p}$ are product- and consumer-specific revenues (€/t), C_i^{est} are establishment costs per hectare, C_i^{opp} are annual opportunity costs per hectare, and C_i^{harv} are harvest and refitting costs per hectare [13]. Transport costs are proportional to flow and distance with rates $c^{\text{tr-raw}}$ and $c^{\text{tr-pre}}$ for raw and pre-treated biomass, respectively, and storage incurs linear holding costs c_j^{stor} per tonne.

363 4.4. Key constraints

364 The path connectivity constraint ensures that each harvest period at a site has exactly
 365 one predecessor and one successor in the age-lag graph, effectively forming a path over
 366 $\mathcal{T}^+$:

$$\sum_{s=0:(s,t) \in \mathcal{S}}^t z_{ist} = \sum_{u=t+1:(t,u) \in \mathcal{S}}^{T_{\max}+1} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}.$$

367 Establishment is restricted to at most one time per site:

$$\sum_{t \in \mathcal{T}^+} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I}.$$

368 Biomass yield is linked to age-lag decisions through the allometric yield coefficients:

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\text{harv}}.$$

369 Harvest quantities bound outbound flows from each site:

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i, p, t \in \mathcal{T}^{\text{harv}}.$$

370 At storage hubs, inventories obey standard balance equations with initial inventory
 371 zero:

$$S_{jpt} = S_{jp,t-1} + \sum_i X_{ijpt} - \sum_{k,p' \geq p} X_{jkpp't} \quad \forall j, p, t \in \mathcal{T}^{\text{harv}}.$$

372 Storage and processing capacities are enforced via:

$$\sum_p S_{jpt} \leq \text{CAP}_j^{\text{stor}} \quad \forall j, t, \quad \sum_{i,p} X_{ijpt} \leq \text{CAP}_j^{\text{proc}} \quad \forall j, t.$$

373 Demand satisfaction with cascades is modelled as described above, where higher-
 374 quality products can satisfy lower-quality demands up to maximum demand levels $D_{kpt}^{\max}$.

375 5. Case study and computational experiment plan

376 The proposed MILP model is intended to support the design of agroforestry biomass
 377 supply chains for cleaner production in specific regional contexts. In a forthcoming case
 378 study, we focus on poplar-based alley-cropping systems in a temperate European setting,
 379 informed by empirical data from SRAFS trials in Southern Germany and SRC plantations
 380 in Central Europe [7, 15, 13].

381 We envisage the following steps and experimental scenarios for computational analy-
382 sis:

- 383 1. **Parameterisation of allometric yield profiles:** Use species- and site-specific
384 allometric models and observed growth trajectories to construct age-dependent
385 yield coefficients η_{pa} for chemical, pulp, and energy product classes at representative
386 agroforestry sites. Where necessary, scenario ranges will be defined to capture
387 uncertainty in MAI and CAI due to climate variability and management differences
388 [6, 7, 15].
- 389 2. **Baseline optimisation:** Solve the MILP model for a baseline configuration with
390 fixed prices, capacities, and policy settings, obtaining optimal establishment timing,
391 harvest schedules, and supply chain structure (locations, flows, inventories).
- 392 3. **Rotation length scenarios:** Vary the age-lag bounds $A^{\min}$ and $A^{\max}$ to represent
393 alternative rotation strategies (e.g., 3–5 years vs. 8–10 years) and analyse their
394 impacts on profitability, product mix, and land use. This will shed light on trade-
395 offs between shorter rotations favouring energy chips and longer rotations enabling
396 higher shares of material products [15, 13].
- 397 4. **Product price and cascading scenarios:** Explore scenarios with different rel-
398 ative prices for chemical, pulp, and energy products, reflecting changes in market
399 demand and policy incentives (e.g., bioenergy subsidies, green chemicals markets).
400 Assess how the product cascade hierarchy and yield quality structure influence the
401 marginal value of species and rotation decisions [2, 8].
- 402 5. **Infrastructure and logistics scenarios:** Evaluate the effect of alternative stor-
403 age and processing network configurations (e.g., decentralised vs. centralised hubs,
404 different capacity levels, transport cost parameters) on the optimal spatial deploy-
405 ment of agroforestry sites and biomass flows [2, 3, 8].
- 406 6. **Sensitivity to allometric and growth uncertainty:** Conduct parametric sen-
407 sitivity and, where computationally feasible, stochastic analyses to quantify the
408 impact of uncertainty in allometric parameters and growth responses (e.g., under
409 drought or management differences) on optimal decisions and system robustness
410 [7, 15].
- 411 7. **Cleaner production indicators:** For selected solutions, compute derived indica-

tors such as biomass cascading index (share of biomass used for material products before energy), utilisation rates of higher-quality fractions, and, where coupled with LCA data, greenhouse gas emission savings relative to fossil benchmarks [5, 16].

6. Conclusions

This paper outlines a research framework that couples allometric biomass growth models with a MILP-based agroforestry biomass supply chain design model featuring age-lag constraints and product cascading. Building on empirical evidence from temperate SRAFS and SRC systems and established biomass supply chain models, the proposed approach aims to capture key interactions between stand-level growth dynamics and network-level design decisions that are central to the cleaner production potential of agroforestry [7, 2, 3]. Future work will implement and calibrate the model for a regional case study, conduct the computational experiments sketched above, and extend the framework towards multi-objective optimisation including environmental indicators such as greenhouse gas emissions and biodiversity proxies.

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